

```

from google.colab import userdata
from groq import Groq
import json
api_key = userdata.get("API_KEY")
client = Groq(api_key=api_key)
product_name = input("Enter a product name: ")
prompt = f"""
Generate information about the product '{product_name}'.

Return ONLY valid JSON with the following fields:
- name
- description
- price
- category

Do not include any explanation or markdown formatting.
"""
response = client.chat.completions.create(
    model="llama-3.3-70b-versatile",
    response_format={"type": "json_object"},
    messages=[
        {
            "role": "user",
            "content": prompt
        }
    ]
)
response_text = response.choices[0].message.content

```

Enter a product name: coffe

```

print("Raw API Response:")
response_text

```

```

Raw API Response:
'{"name": "Coffe",\n  "description": "A stimulating drink made from roasted coffee beans",\n  "price": 2.50,\n  "category": "Beverages"}'

```

```
product_data = json.loads(response_text)
```

```

print("\nPython Dictionary:")
product_data

```

```

Python Dictionary:
{'name': 'Coffe',
 'description': 'A stimulating drink made from roasted coffee beans',
 'price': 2.5,
 'category': 'Beverages'}

```

```

required_fields = [
    "name",
    "description",
    "price",
    "category"
]

try:
    product_data = json.loads(response_text)

    for field in required_fields:
        if field not in product_data:
            raise ValueError(f"Missing field: {field}")

    print("JSON is valid.")
    print(product_data)
except json.JSONDecodeError:
    print("Error: Invalid JSON received.")

except ValueError as e:
    print("Validation Error:", e)

except Exception as e:
    print("Unexpected Error:", e)

```

```
JSON is valid.
{'name': 'Coffe', 'description': 'A stimulating drink made from roasted coffee beans', 'price': 2.5, 'category': 'Beverages'}
```

```
improved_prompt = f"""
Generate details for the product '{product_name}'.
```

Rules:

1. Return ONLY valid JSON.
2. Do not include markdown.
3. Do not include extra text.
4. Include exactly these fields:

```
{{
  "name": "",
  "description": "",
  "price": 0,
  "category": ""
}}
```

5. Price must be a number.

```
"""
```

```
response = client.chat.completions.create(
    model="llama-3.3-70b-versatile",
    response_format={"type": "json_object"},
    messages=[
        {
            "role": "user",
            "content": improved_prompt
        }
    ]
)
```

```
improved_response = response.choices[0].message.content
improved_response
```

```
'{\n  "name": "Coffe",\n  "description": "Freshly roasted arabica coffee beans",\n  "price": 9.99,\n  "category": "Beverage"\n}'
```

```
product_data = json.loads(improved_response)
print("\nImproved JSON:")
product_data
```

```
Improved JSON:
{'name': 'Coffe',
 'description': 'Freshly roasted arabica coffee beans',
 'price': 9.99,
 'category': 'Beverage'}
```